

Network Security

Theory Lesson Nº1

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Outline

- Course presentation
- Professor
- Assessment
- Theory classes
- Practical classes
- Class assignments and Project

Network Security

- **Theory and Practical classes:**

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Opening hours*
Monday: 9:00-12:00
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What is the technology of the Future?

- Genetics and implants
- Robotics
- AI
- Communications and Internet
- Personal Assistants
- Quantum computing
- IoT

  - Security and Privacy

Internet Trends

- Broadband Internet is becoming more widely available
- The cost of connecting is decreasing
- More devices are being created with Wi-Fi capabilities and sensors built into them
- Technology costs are going down
- Smartphone penetration is huge
  - *IoT will forever change our personal and professional lives*

The Internet
the Past, the Present and the Future



Computer Networks

Technologies of Communication Systems

Network Security

Planning of theory classes



1 - Security Fundamentals

2 – Security in Information Systems and Networks

3 - Vulnerabilities

4 - Attacks and detection

5 – Cryptography

6 – Ethical hacking

7 - Solutions and auditing

8 – Privacy

9 - New technologies and trends

Planning of practical classes

1 – Firewalls, Addresses, NAT, VPNs, ...
2 – Cisco Packet Tracer (CCNA, CCNE, ...)
3 – Darkweb
4 – Blockchain
5 – Ethical Hacking
6 – Privacy
7 – Federated Learning
8 – Project support
9 – Evaluations

Network Security
Assessment*

- **Exam** — **8 points**
  Open book exam with 1 sheet
- **Project 1** — **1+2 points**
- **Project 2** — **1+2 points**
- **Lab assignments** — **3 points**
  3 assignments (1 point/assignment)
- BATINA platform — **3 points**
  12*0,25 points
- Extra: Facebook group — 0.2 **points**
  Network Protocols and Technologies

Facebook page



Important rules

- Pedagogical code – Plagiarism!
- The submission of the class assignments and the project must be done on the dates established
- Students with student worker status will be able to submit the project and class assignments until the last class
- There are no faults

Bibliography

Lawrence Gordon and Martin Loeb, “Managing Cybersecurity Resources: A Cost-Benefit Analysis”, Mcgraw-Hill, 2005

F. Boavida, M. Bernardes, "Introdução à Criptografia", FCA editora, 2019

A. Zúquete, “Segurança em Redes Informáticas”, FCA Editora, 2021

Chris Brendan, “Mastering Network Security”, Sybil Network Press, 1998

J. Kurose and K. Ross “Computer Networking – a top-down approach featuring the Internet” Addison-Wesley, 2005.

Manual do Packet Tracer

Kahoot